

Lennard-Jones vs. spring potential

a. Plot the LJ and spring potential

Note:

- σ and ϵ are the constants for length and energy, so I measure lengths in units of σ and energies in units of ϵ . That means $\sigma = \epsilon = 1$ in the code.
- The minimum of LJ is at $r_0 = 2^{1/6} \sigma$, with $V(r_0) = -\epsilon$ (derived in b).
- For the spring I use $V = 1/2 k (r - r_0)^2 - \epsilon$, so its minimum is at the same place as the LJ minimum. Therefore, I can compare them directly.

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In [1]: import numpy as np  
import matplotlib.pyplot as plt
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In [2]: # Lennard-Jones potential:  $V_{LJ}(r) = 4\epsilon\sigma^{12}/r^{12} - \epsilon\sigma^6/r^6$ 
#  $\sigma$  and  $\epsilon$  are the units of length and energy, so  $\sigma = 1$  and  $\epsilon = 1$ 
# Minimum at  $r_0 = 2^{1/6}\sigma$ , where  $V = -\epsilon$  (derived in b)
sigma, eps = 1, 1
r0 = 2**(1/6)*sigma

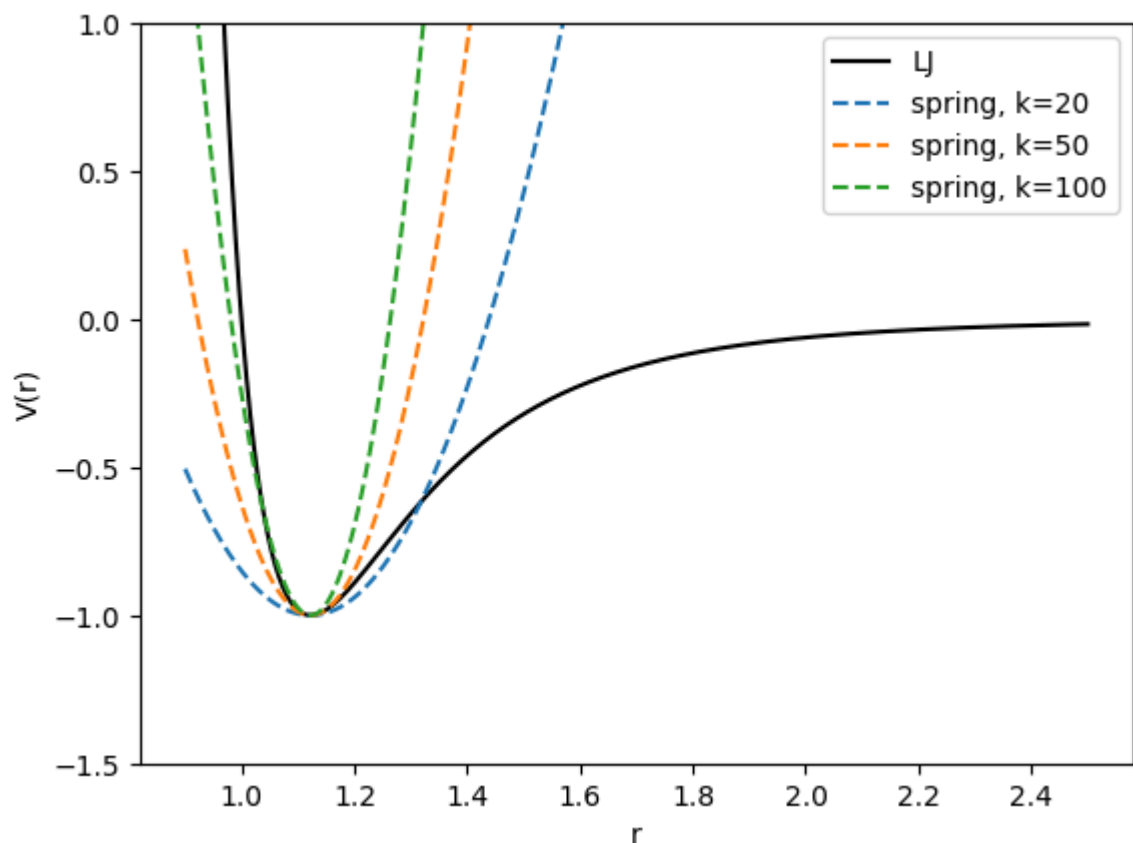
# Spring potential:  $V_{spring}(r) = 1/2*k*(r - r_0)^2 - \epsilon$ 
# I shift it down by eps so its minimum is at the same point as the LJ
# minimum (that is why there is a -eps, just in case you are having a bad day)
# I will try 3 different spring constants: k = 20, 50, 100

# r goes from 0.9 to 2.5 (below 0.9 the LJ potential goes up very fast and
# ruins the plot)
r = np.linspace(0.9, 2.5, 500) #500 points between 0.9 and 2.5

V_LJ = 4*eps*((sigma/r)**12 - (sigma/r)**6)

# Plot LJ (black line) and the springs (dashed lines) in the same figure
fig, ax = plt.subplots()
ax.plot(r, V_LJ, 'k', label='LJ')
for k in [20, 50, 100]:
    ax.plot(r, 0.5*k*(r - r0)**2 - eps, '--', label=f'spring, k={k}') #one
    #dashed line for each k
ax.set_ylim(-1.5, 1) #zoom in around the minimum, otherwise the steep parts
    #take over the plot
ax.set_xlabel('r'); ax.set_ylabel('V(r)'); ax.legend()
plt.show()

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Note:

- Both potentials have a minimum. There, $F = -dV/dr = 0$, so it is a stable equilibrium.
- If a particle is pushed a bit away from the minimum, the force pushes it back. Therefore, it oscillates around the minimum.
- Close to the minimum the LJ potential looks like a parabola, just like the spring. For k around 50 they almost overlap.
- Further away they are different. LJ is very steep for $r < r_0$ and flattens to 0 for large r , so the bond can break. The spring keeps growing.

b. Small oscillations

Note:

- The particles are already bonded, so they sit at the minimum r_0 . A small push moves them a bit away, so r stays close to r_0 .
- When r is close to r_0 , I can Taylor expand $V(r)$ around r_0 :

$$V(r) = V(r_0) + V'(r_0)(r - r_0) + \frac{1}{2}V''(r_0)(r - r_0)^2 + \frac{1}{6}V'''(r_0)(r - r_0)^3 + \dots$$

- Because $r - r_0$ is small, $(r - r_0)^3$ and higher powers are much smaller than $(r - r_0)^2$. Therefore, I only keep the terms up to $(r - r_0)^2$.
- So I need three things: $V(r_0)$, $V'(r_0)$ and $V''(r_0)$.

Step 1: the first derivative, and where the minimum is

- I write V with powers of r : $V(r) = 4\epsilon(\sigma^{12} r^{-12} - \sigma^6 r^{-6})$. Then

$$V'(r) = 4\epsilon \left(-\frac{12\sigma^{12}}{r^{13}} + \frac{6\sigma^6}{r^7} \right)$$

- At the minimum the slope is zero, so $V'(r_0) = 0$. This means $12\sigma^{12}/r_0^{13} = 6\sigma^6/r_0^7$. Multiplying both sides by $r_0^{13}/(6\sigma^6)$ gives $r_0^6 = 2\sigma^6$, so $r_0 = 2^{1/6}\sigma$.
- Therefore, the $(r - r_0)$ term in the Taylor expansion is zero.

Step 2: the numbers I need at r_0

- From step 1, $r_0^6 = 2\sigma^6$. So $(\sigma/r_0)^6 = 1/2$ and $(\sigma/r_0)^{12} = 1/4$.
- (In case that was too fast: $r_0 = 2^{1/6}\sigma$, so $\sigma/r_0 = 2^{-1/6}$. Then $(2^{-1/6})^6 = 2^{(-1/6 \cdot 6)} = 2^{-1} = 1/2$, because $(a^b)^c = a^{(b \cdot c)}$. Why? Because $(a^b)^c$ means a^b multiplied by itself c times, and when you multiply powers of the same number you add the exponents, so you get $a^{(b + b + \dots + b)}$ with c times b , and $b + b + \dots + b$ (c times) is $b \cdot c$, which is literally what multiplication means. For a fraction like $1/6$ the rule still works, because $2^{1/6}$ is defined as the number that gives 2 when you multiply it by itself 6 times, which was chosen exactly so this rule keeps working. Then $(\sigma/r_0)^{12} = ((\sigma/r_0)^6)^2 = (1/2)^2 = 1/4$, because $12 = 6 \cdot 2$ (I checked) and I use the same rule I just explained, just in case you are having a bad day.)

Step 3: $V(r_0)$

$$V(r_0) = 4\epsilon \left(\frac{1}{4} - \frac{1}{2} \right) = -\epsilon$$

Step 4: the second derivative

- I differentiate $V'(r)$ once more ($-12 \cdot (-13) = 156$ and $6 \cdot (-7) = -42$):

$$V''(r) = 4\epsilon \left(\frac{156\sigma^{12}}{r^{14}} - \frac{42\sigma^6}{r^8} \right)$$

- At r_0 I use $\sigma^{12}/r_0^{14} = (\sigma/r_0)^{12} \cdot 1/r_0^2 = 1/(4 r_0^2)$ and $\sigma^6/r_0^8 = (\sigma/r_0)^6 \cdot 1/r_0^2 = 1/(2 r_0^2)$:

$$V''(r_0) = \frac{4\epsilon}{r_0^2} \left(\frac{156}{4} - \frac{42}{2} \right) = \frac{4\epsilon}{r_0^2} (39 - 21) = \frac{72\epsilon}{r_0^2} = \frac{72\epsilon}{2^{1/3}\sigma^2}$$

- The last step uses $r_0^2 = (2^{1/6} \sigma)^2 = 2^{1/3} \sigma^2$.

Result: putting steps 1 to 4 into the Taylor expansion,

$$V(r) \approx -\epsilon + \frac{1}{2} \cdot \frac{72\epsilon}{2^{1/3}\sigma^2} (r - r_0)^2 = -\epsilon + \frac{36\epsilon}{2^{1/3}\sigma^2} (r - r_0)^2$$

Therefore, close to the minimum the LJ potential is a parabola, which is what we saw in the plot in a.

c. Spring constant

Note:

- For a spring (harmonic oscillator) with rest length r_0 , the potential is $V = 1/2 k (r - r_0)^2$ and the force is $F = -dV/dr = -k (r - r_0)$ (Hooke's law).
- From b, the LJ potential near the minimum is

$$V(r) \approx -\epsilon + \frac{1}{2} V''(r_0) (r - r_0)^2$$

- The constant $-\epsilon$ does not change the force, because the derivative of a constant is 0 (just in case). So only the $(r - r_0)^2$ term matters.
- Comparing the two term by term: $1/2 k = 1/2 V''(r_0)$. Therefore,

$$k = V''(r_0) = \frac{72\epsilon}{2^{1/3}\sigma^2} \approx 57.1 \frac{\epsilon}{\sigma^2}$$

- With $\sigma = \epsilon = 1$ this is $k \approx 57.1$. This matches the plot in a, where $k = 50$ already almost overlapped with LJ near the minimum.

d. Two bonded masses

Note:

- The two masses are connected by a spring with the k from c. In equilibrium (first picture) the spring has its rest length, so there is no force.
- In the second picture m_1 has moved r_1 to the right and m_2 has moved r_2 to the left. They moved towards each other, so the spring is compressed by $r_1 + r_2$.

- A compressed spring pushes both masses back with Hooke's law, with force $k(r_1 + r_2)$. m_1 is pushed to the left (against r_1) and m_2 to the right (against r_2).

Force equations (Newton's second law, $F = ma$, for each mass):

$$m_1 \ddot{r}_1 = -k(r_1 + r_2)$$

$$m_2 \ddot{r}_2 = -k(r_1 + r_2)$$

- The minus sign is there because the force points opposite to the displacement (just in case you are having a bad day).
- Both equations only depend on $r_1 + r_2$, so I turn them into one equation for $x = r_1 + r_2$. First I divide each equation by its mass:

$$\ddot{r}_1 = -\frac{k}{m_1}(r_1 + r_2), \quad \ddot{r}_2 = -\frac{k}{m_2}(r_1 + r_2)$$

- Then I add them:

$$\ddot{r}_1 + \ddot{r}_2 = -k \left(\frac{1}{m_1} + \frac{1}{m_2} \right) (r_1 + r_2)$$

- With $x = r_1 + r_2$ and the reduced mass μ , defined by $1/\mu = 1/m_1 + 1/m_2$ (so $\mu = m_1 m_2 / (m_1 + m_2)$), this becomes

$$\ddot{x} = -\frac{k}{\mu} x$$

- x is how much the distance between the particles changed, so it is the same as $r - r_0$ in b and c (up to a sign, which does not matter for a spring).
- This is the equation of a harmonic oscillator. Its solution is $x(t) = A \cos(\omega t + \phi)$. Putting it in gives $\ddot{x} = -\omega^2 A \cos(\omega t + \phi) = -\omega^2 x$, so $\omega^2 = k/\mu$.

Therefore, the particles oscillate with angular frequency ω and frequency f :

$$\omega = \sqrt{\frac{k}{\mu}}, \quad f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

With the LJ spring constant from c:

$$\omega = \sqrt{\frac{72\varepsilon}{2^{1/3}\sigma^2\mu}} \approx 7.56 \sqrt{\frac{\varepsilon}{\sigma^2\mu}}$$

- Check: if m_2 is very heavy (like a wall), then $\mu \approx m_1$ and $\omega = \sqrt{k/m_1}$, which is the normal mass on a spring.